산악 지역 분산형 급수 인프라의 공간적 적합성 평가 및 입지 선정: 네팔 카르날리주 사례 연구

Spatial Multi-Criteria Suitability Assessment and Water Supply System Siting for Decentralized Infrastructure in Complex Mountainous Terrain: A Case Study in Karnali Province, Nepal

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연구배경

- ❑ Springs are declining across the Nepalese Himalayas, and in Rekcha (Chaukune Municipality, Karnali Province), water shortage cut the village from about 180 to 70 households.
- ❑ Steep slopes, large elevation differences, long distance to rivers and landslide susceptibility limit where water infrastructure can be built
- ❑ Regional plans that use broad centroid points can miss local elevation differences that decide whether a source can supply a village
- ❑ Therefore, this study develops a two-level GIS framework: Level 1 maps suitability with GIS-AHP and assigns preliminary technologies by rule, while Level 2 assesses and proposes a preliminary two-stage solar pumping and gravity design

연구방법

I. Study area and data

1. Study area and tools

- ❑ Study area: Karnali Province, Nepal (27,984 km²; 10 districts, 79 municipalities) with a case study at Rekcha, Ward 5, Chaukune Rural Municipality (Fig. 1)
- ❑ Spatial unit: 30 m raster cells (WGS 84 / UTM 44N); tools: QGIS, GRASS GIS, SAGA

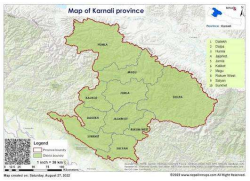


Fig.1: Karnali province and its ten districts

Table 1. AHP criteria classification and weight

Criterion	Weights	Classification (1/2/3)
River Distance	42.4%	>15km/3-15km/0-3km
Slope	22.7%	0-8%/ >57%> 8-57%
Elevation	22.7%	<500m/500m-3500m/>3 500m
Land Cover	12.2%	Barren/ Cropland/ Forest

2. Data and Criteria

- ❑ Four criteria were scored 1 (Low) to 3 (High); landslide susceptibility was used as an exclusion mask, not a criterion

II. Level1: Provincial suitability and technology assignment

1. GIS-AHP suitability model

- ❑ Criteria weights come from AHP pairwise comparison (consistency ratio = 0.4%)
- ❑ Weighted overlay: Score = 0.424 River + 0.227 Slope + 0.227 Elevation + 0.122 Land cover
- ❑ NASA landslide class was excluded and class 4 flagged for geotechnical checks
- ❑ Scores were grouped into Low, Moderate, High and Very High suitability

2. Technology assignment (Table 2)

- ❑ Each 30 m cell was classified into four technology options using suitability (S), elevation (E), and river distance (R)
- ❑ Rules provide preliminary spatial guidance, not final engineering or construction decisions

Table 2. Rules for preliminary technology assignment

Class	Technology	S	E (m)	R (m)
1	Gravity-fed	>=3	<=1500	<=3000
2	SPWPS	>=3	1500-3500	<=5000
2	SPWPS	>=3	<=1500	3000-5000
3	Point source	=2	<=3500	any
4	Exclusion	=1	any	any
4	Exclusion	any	>3500	any

III. Level 2: Rekcha settlement-scale assessment

1. Terrain and elevation check

- ❑ A 30m DEM was clipped to 3.5km buffer, sinks were filled and streams extracted
- ❑ Esri imagery was used to check the settlement location visually
- ❑ Head difference: ΔH = H_intake − H_village

2. Preliminary design

- ❑ Two stages (solar rising main, then gravity main) designed with reference to Nepal DWSSM static-head limits (15–60m)
- ❑ Demand: 455 residents × 45 L/day = 20.5 m³/day; pump flow 3.5 m³/h

연구결과

I. Provincial suitability and technology assignment

1. Criterion maps and landslide screening

- ❑ Criterion maps show where river distance, slope, elevation and land cover score highest
- ❑ Most of the province has high landslide susceptibility (mean 4.38 of 5), so class 5 cells were excluded and class 4 flagged

연구결과

2. Provincial suitablity

- ❑ High and Very High suitability follows mid-hill river corridors in Dailekh, Surkhet, Salyan and Jajarkot, home to about 66% of the population covered
- ❑ Chaukune, the case municipality, has 84.41% High or Very High suitability

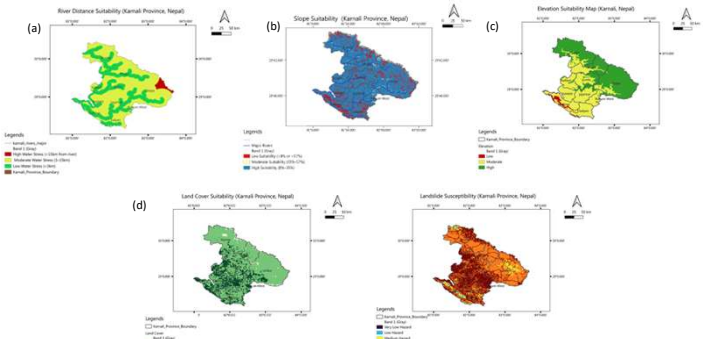


Fig. 2: criterion maps, (a) river distance, (b) slope, (c) elevation, (d) land cover

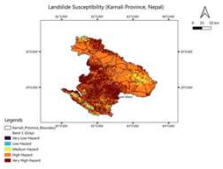


Fig. 3: landslide susceptibility map (NASA, 1–5)

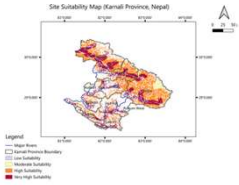


Fig. 4: provincial suitability map (Low to Very High)

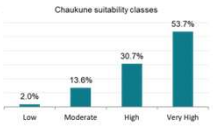


Fig. 5: Chaukune class shares

3. Technology assignment

- ❑ Solar pumping covers 29.06% of the valid area, gravity 13.35%, point sources 3.27% and 54.32% was excluded
- ❑ Solar pumping covers about twice the area of gravity

III. Rekcha settlement-scale assessment

1. Satellite verification

- ❑ Rekcha sits at 1,312 m, not 1,240 m as the regional centroid suggested
- ❑ ΔH = 1,278 − 1,312 = −34 m, so gravity supply from this intake is not feasible without lifting

2. Preliminary two-stage design

- ❑ Stage 1: an 82 m solar lift through a 771 m rising main to a ridge reservoir at 1,360 m
- ❑ Stage 2: a 566 m gravity main with 48 m of static head, within the DWSSM range
- ❑ Total dynamic head 92m, pump power 0.88kW, PV array 1.8–2.2 kWp

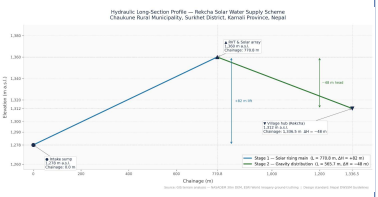


Fig.7. Hydraulic long-section profile (Elevation vs. Chainage), Rekcha SPWPS

결론

- ❑ This study developed a two-level GIS framework that links province-wide screening to a settlement-scale check for decentralized water supply planning
- ❑ Suitability follows mid-hill river corridors, and Chaukune, the case municipality, has 84.41% High or Very High suitability. The preliminary technology assignment placed 29.06% of valid area in solar pumping and 13.35% in the gravity-fed class
- ❑ At Rekcha, the settlement lies 34 m above the intake, so a two-stage solar and gravity design is proposed (82 m solar lift, 48 m static head)
- ❑ Regional suitability alone cannot confirm hydraulic feasibility, settlement-scale assessment is essential
- ❑ Limitations: Remote data and one settlement limit the assessment. Field verification and flow measurements are needed before detailed design